

Herald



Herald — Fixed

Field	Value
Revision	19
Created	2026-05-20
Last modified	2026-05-30
Status	active
Status summary	r19 (v1.0.0 closure-migration) atomically lands (§11.4.19) 51 code-verified HRDs Issues→Fixed after a re-verification audit (grep/read the impl + run the test + confirm the matching e2e invariant + confirm the docs/qa/<run-id>/ dir; nothing migrated on self-reported status alone). The 51: the 7 flavor constitution bindings HRD-019..025 (<flavor>/internal/bindings Pipelines; 7/7 bindings test suites PASS - race; e2e E90-E96; per-HRD docs/qa/); the entire §43 command catalogue HRD-029..056 (30 commands wired into Cobra roots via registerGitOps/registerSysOps/etc., cmd-test suites PASS, e2e E97-E122, per-HRD docs/qa/); the Batch-D commons_infra.pgxTaskRepository ops HRD-085..089 (all 15 methods real impls not ErrUnsupported, migration 000013_task_resource_snapshots, e2e E123, docs/qa/HRD-085-089-*); Wave 7 HRD-116/118/119/120/121 (qaherald Slack MessengerClient + e2e E124-E131 + paired §1.1 gate tests/test_wave7_mutation_meta.sh + multi-channel §11.4.85 stress/chaos + MESSENGER_CHANNELS.md guide; all hermetic tests PASS); ATMOSphere features HRD-148/149/151/152/153/154 (commons_workable+commons_watch build+test PASS, change→CloudEvent→Runner bridge + inbound action registry + pherald watch daemon + quiet-hours resolver, e2e E139 TestRunWatch_EndToEndOutbound PASS hermetic). NOT migrated (evidence-gated, left in Issues.md): HRD-115 (Slack channel adapter — operator-gated on LIVE docs/qa/HRD-115-LIVE-*/ round-trip evidence per §107.x; only hermetic transcript exists, E127 LIVE-Slack SKIP-with-reason — closing now would be a §107 PASS-bluff); HRD-117 (spec V3 doc-only — §106); HRD-131/150/155/156/157/158 (externally gated); HRD-146/147/081/008/015 (not-started/upstream/operator). Prior r18 atomically lands (§11.4.19) HRD-090 dead-letter handling Completed→Fixed — commons_infra.pgxTaskRepository.MoveToDeadLetter (was the last Batch-D ErrUnsupported stub): migration

Field	Value
000014_dead_letter_tasks	+ tx-atomic JSONB-snapshot move + terminal-mark; new 13th OPERATIONAL constitution event class ClassQueueDeadLetter + QueueEvent + DeadLetter() emit (wired nil-tolerantly; governance §42.2 set unchanged). Four-layer closure: 3 unit + real-PG integration (3 sinks) + §11.4.85 stress/chaos (50 concurrent race-clean, atomic under fault) + paired §1.1 mutation gate (4/0) + e2e E138; commits 549261d+9c765f6; evidence docs/qa/HRD-090-20260529T052228Z/ + docs/qa/HRD-090-stress-chaos-20260529T054500Z/. Governance-plane subscriber follow-up = HRD-147 (Issues.md).** Prior r17 atomically lands (§11.4.19) the completed Wave 8 Track B (MTPProto user-account harness for §11.4.98 full-automation testing) — HRD-139 (MTPProto Client live wiring + qaherald mtproto CLI), HRD-140 (Subscribe roundtrip), HRD-141 (Wave 6 closed loop), HRD-142 (Wave 6.5 lifecycle) all New→Fixed (these HRDs were tracked directly in the TaskList rather than Issues.md and graduate directly into Fixed.md per §11.4.19 the moment closure landed).** Wave 8 Track B closure proves §11.4.98 (full-automation anti-bluff mandate) end-to-end: 3 MTPProto-driven live tests (E135 Subscribe / E136 Wave 6 closed loop / E137 Wave 6.5 lifecycle) all PASSING LIVE under §11.4.98-pure interpretation, captured under docs/qa/HRD-LIVE-MTPROTO-20260528T125321Z/. Three legacy NON-COMPLIANT tests graduate to §11.4.90 Obsolete in Issues.md r36 with Obsolete-Details cite of §11.4.98 + §108.m superseding-mandate: TestSubscribe_LiveBotAPI, tests/test_wave6_live_loop.sh, lifecycle --manual mode. v0.6.0 tag (844f340 annotated) was already cut on d843ddf covering Wave 8 Track B closure + the §11.4.98+§11.4.99 anchoring + 9-Helix-submodule §11.4.98+§11.4.99 propagation. Prior r16 atomically migrates (§11.4.19) the completed Tgram-completeness wave — HRD-133/134/135/136/137/138 (all 6 Tgram-completeness HRDs) Issues→Fixed.** Wave A: HRD-135 lifecycle handlers (fe2a7c8 — OnEdited / OnMyChatMember / serviceHandler / ErrBotKicked sentinel) + HRD-136 webhook secret_token validator (6d7d543 — crypto/subtle.ConstantTimeCompare, 401-no-leak posture) + HRD-138 operator deep-dive guide (ff9b9c0 — 810-line docs/guides/TELEGRAM.md, 11 §-sections, 3 siblings). Wave B: HRD-133 token-redaction symmetry (8162d14 — sanitizeTgramError wraps 6 error-return sites in Send/SendReply) + HRD-134 429 retry_after honored (8162d14 combined — withRetry429 wraps all 3 telebot bot . Send sites, telebot.FloodError typed extraction, ctx-aware sleep, 3-attempt 60s-clamp). Wave C: HRD-137 stress+chaos suite (this commit — TestTgram_Stress_MultiRecipientFanOut 20×20=400 calls/20 chats deterministic + TestTgram_Chaos_GetUpdatesPollerResilience SKIP→PASS conversion via one-line subscribe.go baseURL seam fix; 8-phase fault matrix deterministic across -race -count=3 — every iteration IDENTICAL recovery=6 fault(500=12 hang=8 close=3 ok=6) goroutines 3→4). Full tgram suite 26/26 PASS, 0 SKIP post-Wave-C. All 6 HRDs carry docs/qa/HRD-NNN-20260528T*/transcript.txt evidence per §107.x. Prior r15 atomically migrated (§11.4.19) the completed GAP-3 §11.4.85 stress+chaos suite — HRD-122..HRD-130 (Task→Completed) — and the HRD-132 Runner idempotency-window fix (Bug→Fixed): 10 rows Issues→Fixed.** GAP-3 commits 02d272d→d2a60d3 (HRD-122 commons/stresschaos/ scaffold, HRD-123 Gin /v1/events, HRD-124 /v1/compliance+/v1/safety_state, HRD-125 Runner — surfaced HRD-132, HRD-126 pherald listen inbound, HRD-127 claude_code dispatch, HRD-128 container/resource §12.6-safe, HRD-129 e2e E81-E88, HRD-130 paired

Field	Value
	§1.1 mutation gate); HRD-132 fix 354b883 (claim-before-dispatch at Runner Stage 2 — INSERT events_processed ... ON CONFLICT DO NOTHING is the authoritative exactly-once dispatch gate; latent runner.go:132 CachedRcpt.WasReplay race closed; HRD-125 assertion upgraded to sends==1) + wave3-M4-retire 4f45f26. Full scripts/e2e_bluff_hunt.sh 78 PASS / 0 FAIL; HRD-130 gate 6 PASS / 0 FAIL (5 load-bearing mutations); HRD-132 evidence docs/qa/HRD-125-stress-chaos-2026-05-27T131251-gap3/ (dispatch_exactly_once=1). HRD-131 (SQLite SSoT migration Phase 1) remains OPEN. Prior r14 captured Wave 6 (pherald inbound runtime + Claude Code headless bridge — code-doc closure). HRD-100 closed atomically with 12 Wave-6 commit SHAs + 8 new e2e invariants E63-E70 (currently SKIP-with-documented-reason — converting to PASS once operator runs the live closed-loop at T10b and commits docs/qa/HRD-100/). Tag v0.4.0 DEFERRED to T13b (operator-supplied live evidence per §107.x — code-doc closure ships in this commit; runtime evidence lands separately).** Wave 6 surfaces: new Cobra subcommand pherald listen (Telegram getUpdates long-poll + OnText/OnPhoto/OnDocument/OnVoice + bot self-filter + sha256 content-addressed attachment download); Opus model pin in spawned claude argv (--model claude-opus-4-7); envelope pre-text — verbatim operator wording "We have received new message from our communication channel" preceding the <<<HERALD-DISPATCH-v1>>> structured block; <<<HERALD-REPLY>>> parser with action routing (reply / issue.open / event.emit); tgram.Adapter.SendReply wiring reply_to_message_id; --qa-out-dir JSONL journaling per §107.x. Wave 6 mutation gate (tests/test_wave6_mutation_meta.sh) lands 3 paired hermetic mutations (M1 blank bot self-filter → echo-loop signature; M2 Opus → Sonnet model swap → E67 FAIL; M3 drop opts.ReplyTo in SendReply → E68 FAIL). Prior r13 captured Wave 4b (TOON application / toon content negotiation; tag v0.3.0). r12 captured Wave 4a (HTTP/3 + Brotli + Alt-Svc + TLS 1.3; tag v0.2.0). r11 closed HRD-016 (Wave 3b pherald Runner live). r10 closed HRD-011 (Telegram live).
Issues	see Issues.md — r43 (Issues.md) captures this r19 §11.4.19 atomic migration of 51 code-verified HRDs (flavor bindings HRD-019..025, §43 catalogue HRD-029..056, Batch-D pgx HRD-085..089, Wave 7 HRD-116/118/119/120/121, ATMOSphere features HRD-148/149/151/152/153/154); 17 items remain open in Issues.md.
Issues summary	17 still open in Issues.md: HRD-008, HRD-015, HRD-081 (operator/upstream); HRD-115 (Slack adapter — operator-gated on LIVE round-trip evidence per §107.x); HRD-117 (spec doc-only); HRD-131, HRD-150, HRD-155, HRD-156, HRD-157, HRD-158 (ATMOSphere/SSoT externally gated); HRD-146, HRD-147 (dispatch-hardening / governance-subscriber follow-ups); HRD-143/144/145 (Obsolete).
Fixed	HRD-001..HRD-007, HRD-009, HRD-009b, HRD-010, HRD-011, HRD-012, HRD-013, HRD-014, HRD-016, HRD-017, HRD-018, HRD-019..HRD-025, HRD-026, HRD-027, HRD-028, HRD-029..HRD-056, HRD-080, HRD-085..HRD-089, HRD-090, HRD-092, HRD-093, HRD-094, HRD-095, HRD-096, HRD-097, HRD-098, HRD-099, HRD-100, HRD-110, HRD-111, HRD-112, HRD-113, HRD-114, HRD-116, HRD-118, HRD-119, HRD-120, HRD-121, HRD-122, HRD-123, HRD-124, HRD-125, HRD-126, HRD-127, HRD-128, HRD-129, HRD-130, HRD-132, HRD-133, HRD-134, HRD-135, HRD-136, HRD-137, HRD-138, HRD-139, HRD-140, HRD-141, HRD-142, HRD-148, HRD-149, HRD-151, HRD-152, HRD-153, HRD-154

Field	Value
Fixed summary	v1.0.0 closure-migration (r19 — 51 HRDs: flavor constitution bindings HRD-019..025, §43 command catalogue HRD-029..056, Batch-D pgxTaskRepository ops HRD-085..089, Wave 7 multi-channel HRD-116/118/119/120/121, ATMOSphere workable-items features HRD-148/149/151/152/153/154 — each re-verified against committed code + passing tests + e2e invariant + docs/qa before migration). Wave 8 Track B (HRD-139/140/141/142 — MTPProto user-account harness proves §11.4.98 full-automation end-to-end with 3 live MTPProto-driven tests PASSING LIVE 2026-05-28; v0.6.0 tag covers the closure); Tgram-completeness wave (HRD-133/134/135/136/137/138 — 6 HRDs across 3 sub-waves A/B/C, full tgram suite 26/26 PASS 0 SKIP); GAP-3 §11.4.85 stress+chaos suite (HRD-122..HRD-130 Completed) + HRD-132 Runner idempotency-window fix (claim-before-dispatch, exactly-once dispatch <code>sends==1</code>); spec V1→V3 r9; Go module foundation + Foundation M1/M2/M3; HRD-010 commons_storage live wiring; HRD-011 Telegram live (live message_id=5 evidence 2026-05-22); HRD-012 Claude Code dispatcher live; HRD-016 pherald /v1/events Runner wiring (Wave 3b live — 7-stage pipeline + e2e E37-E42); universal §11.4.73 + §11.4.74 mandates propagated; I6 gate refined; Wave 2 flavor scaffolds (HRD-092..097) closed atomically; Wave 3a substrate (HRD-099 commons_auth JWT verifier) + 2 live REST routes (HRD-028, HRD-098) closed atomically. Continuation see CONTINUATION.md.

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ID	Type	Criticality	Title
HRD-019	task	high	cherald constitution bindings — cherald/internal/bindings Pipeline registers the 15 cherald-owned §42.3 rules as concrete commons_constitution.Evaluator impls routed through 5 event classes (.policy.violation/.gate.failed/.gate.recovered/.credentials) over the Batch-A emit+persist+audit foundation; POST /v1/compliance/catalogue surface drives detect→emit→persist→audit, visible on the existing GET /v1/compliance/catalogue surface. Vertical-slice bespoke detectors landed (§11.4.29 naming, §11.4.100 doc-revision-header, §11.4.44 doc-revision-header, §11.4.18 script-docs, §11.4.73 spec-drift, §11.4.74 remaining 25 rules use the shared marker-check mechanism (verdict decided up on catalogue command / sweep) with the full emit-routing framework in place — the remaining bespoke detector is per-rule follow-up. sherald host-safety + repo-safety constitution bindings — sherald/internal/bindings Pipeline registers the 15 sherald-owned §42.3 host/repo-safety rules (§11.4.24/.26/.32/.36/.41/.47/.71, §12.1/.2/.3/.6) as concrete commons_constitution.Evaluator impls routed through 6 SAFETY event classes (.host.safety.breach/.repo.safety.breach/.gate.recovered/.credentials) over the Batch-A emit+persist+audit foundation, mirroring the proven HRD-019 pattern. The 3 named DETECTION hooks landed as REAL pure classifiers — c detector (§9.1: rm/git-reset/clean without backup → .repo.safety.breach) interceptor (§11.4.41/§9.2: force-push without merge-first OR session-auth → .repo.safety.breach), mem-budget watcher (§12.6: reported used_fraction)
HRD-020	task	high	

ID

Type Criticality Title

			.host.safety.breach) — plus forbidden-host-op (§12.1) + prereq-marker remaining rows. CRITICAL §12-safety: every detector DETECTS+REPORTS string description; NONE performs the destructive op, force-pushes, suspends the allocates memory (the mem-budget watcher reads a fabricated fraction — no real §43 command bodies (HRD-033/046/056) supply the live op-interception feeding bherald CI/test bindings — bherald/internal/bindings Pipeline re bherald-owned §42.3 CI/test gate-result rules (§1, §1.1, §11.4.2/.3/.4/.5/.7/.9/.13/.14/.24/.27/.30/.39/.43/.46/.48–.52/.67) as concrete commons_constitution.Evaluator impls routed through 3 event classes (.gate.failed/.gate.recovered via the build/test gate-result class + .policy.violation + .repo.safety.breach for the shared §11.4.30 row) over the Batch-A emit+persist+audit foundation, mirroring the proven HRD-019/020/021 Pipeline pattern. bherald is the BUILD/CI flavor; it emits CI/test gate-result com that integrate with the consuming project's CI workflow. 3 bespoke pure detector result classification (§1/§11.4.50 — PASS/FAIL/flaky from a recorded gate outcome .gate.failed/.gate.recovered), test-tier-verify (§11.4.27/§40.2 — the matrix; a missing tier FAILs), anti-bluff-PASS detection (§11.4.2/.5 — a gate th WITHOUT a captured-evidence artefact is a §11.4 PASS-bluff → FAIL). Remain shared marker-check mechanism (verdict decided upstream by the §43 command full emit-routing framework in place. rherald release constitution bindings — rherald/internal/binding registers the 4 rherald-owned §42.3 release-lifecycle rules (§4 tag mirroring, §5 multi-format export [shared with cherald], §11.4.38 installable-asset evidence, § retest before release tag) as concrete commons_constitution.Evaluator through 2 event classes (.release.gate.blocked via the existing ReleaseGateBlocked emitter surface for §4/§11.4.38/§11.4.40 + .policy for the shared §5 changelog row; a release recovery maps to the shared .gate companion since the release class family has no distinct "recovered" leaf) over the emit+persist+audit foundation, mirroring the proven HRD-019/020/021 Pipeline the RELEASE flavor. 4 bespoke PURE detectors landed: tag-mirror-parity (§4 - parent but missing on any owned mirror, with_tag<mirrors, → .release.gate.blocked), changelog-conformance (§5 — non-Conventional changelog OR stale multi-format export → .policy.violation), pre-tag-r (§11.4.40 CRITICAL — tag attempted without a green all-8-§40.2-tier retest → .release.gate.blocked), installable-asset-evidence (§11.4.38 — release install check → .release.gate.blocked). CRITICAL §12-safety: every CLASSIFIES a recorded release-op outcome string; NONE tags, pushes, force-downloads, or installs — the live release-op interception is scope-locked to the (HRD-031/032/045). pherald project bindings (§2 commit+push, §3 submodule propagation, §11.4.11/.15/.21/.22/.34/.37/.42/.55/.66/.71/.74) — Batch B: pherald/internal/bindings/ Pipeline + §42.3-owned rule set + 5 bespoke PURE detectors (con discipline / submodule-propagation-order / pre-push-fetch-guard / fetch-before-through the commons_constitution Evaluator+audit foundation; -race -count=3 Live op-interception deferred to §43 HRD-029/030/043/044/049/053. iherald constitution-rule escalation bindings (§11.4.10/10.A credential-leak + §11.4.66 operator-blocked escalation) — Batch B: iherald/internal/b Pipeline routing .credential.leak (forced-critical) + .policy.violation detectors wired through commons_constitution Evaluator+audit; -race -count=3 no-real-secret attestation. Live paging deferred to §43 / HRD-024-paging.
HRD-021	task	middle	
HRD-022	task	middle	
HRD-023	task	middle	
HRD-024	task	middle	
HRD-025	task	low	

ID	Type	Criticality	Title
			§11.4.45 sherald scheduled-audit bindings (§11.4.45 periodic Status.md sweep + daily compliance digest) — Batch B: sherald/internal/bindings/Pipeline .policy.violation + 3 bespoke detectors (status-sweep / digest-cadence) through commons_constitution Evaluator+audit; -race -count=3 green, e2e E96 scheduler deferred to §43 HRD-047.
HRD-029	task	middle	§2 pherald commit-push — single-entripoint locked commit + multi-mirror Batch C: implemented as gitops command (§11.4.74 wrap of git binary); --emit feeds hermetic temp-repo tests -race green
HRD-030	task	middle	§3 pherald submodule propagate — owned-submodule walk in propagate Batch C: implemented as gitops command (§11.4.74 wrap of git binary); --emit binding; hermetic temp-repo tests -race green
HRD-031	task	middle	§4 rherald tag mirror — assert tag exists on every owned submodule — implemented as rherald §43 release command (rherald/cmd/rherald/release_cmds.go) feeding the HRD-022 rherald binding; reuses shared commons hermetic -race green
HRD-032	task	low	§5 rherald changelog generate — Conventional Commits → docs/ <v>.md + multi-format export — Batch C: implemented as rherald §43 release (rherald/cmd/rherald/release_cmds.go) feeding the HRD-022 rherald binding; reuses shared commons/gitops; hermetic -race green
HRD-033	task	high	§9.1 sherald destructive guard <op> — wrap rm/git-reset/git-push prerequisite checks — Batch C: implemented as sherald §43 command (sherard/cmd/sysops_cmds.go) feeding the HRD-020 sherald binding; PRE-FAIL (detects+classifies+exits-nonzero, never performs the op); hermetic -race green; fix so the CLI runs without a JWT secret
HRD-034	task	middle	§9.3 sherald backup-snapshot <target> — hardlinked snapshot hermetic straggler: implemented in sherald/cmd/sherald/sysops_cmds.go (command refuses overwrite, classifies §9.3 SubjectBackup → .gate.recovered); hermetic equality -race green
HRD-035	task	middle	§11.4.2 bherald evidence capture <test_id> — captured-evidence Batch C: implemented as bherald §43 build/CI command (bherald/cmd/bherald/build_cmds.go) feeding the HRD-021 bherald binding; hermetic -race green
HRD-036	task	high	§11.4.10 cherald creds scan — gitleaks/trufflehog integration, emits .credential.leak — Batch C: implemented as cherald §43 verify/check command (cherald/cmd/cherald/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only; hermetic -race green (routed to §11.4.10 credential-leak; findings R secret value printed)
HRD-037	task	low	§11.4.12 cherald docs sync — regen Issues_Summary / Fixed_Summary / Status_Summary — Batch C: implemented as cherald §43 docs-pipeline command (cmd/cherald/docops_cmds.go) feeding the HRD-019 cherald binding; - - apply to regenerate; ships the §107 lazy-verifier fix so the CLI runs without JWT secret; hermetic -race green
HRD-038	task	low	§11.4.18 cherald script-docs check — assert sibling .md for every script Batch C: implemented as cherald §43 verify/check command (cherald/cmd/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only; hermetic (routed to §11.4.60 — §11.4.62 has no cherald rule)
HRD-039	task	low	§11.4.19/.23 cherald fixed align+cherald colorize — Issues/Status/HTML colorizer — Batch C: implemented as cherald §43 docs-pipeline command (cmd/cherald/docops_cmds.go) feeding the HRD-019 cherald binding; - - apply to regenerate; ships the §107 lazy-verifier fix so the CLI runs without JWT secret; hermetic -race green

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HRD-040	task	high	§11.4.26 sherald constitution pull — wrap fetch + rebase + validate .bundle.updated — Batch C: implemented as sherald §43 command (sherald/sysops_cmds.go) feeding the HRD-020 sherald binding; PRE-FLIGHT GATE (detects+classifies+exits-nonzero, never performs the op); hermetic -race green; fix so the CLI runs without a JWT secret
HRD-041	task	middle	§11.4.27 bherald test-tier verify — 8-tier matrix verification — Batch C: implemented as bherald §43 build/CI command (bherald/cmd/bherald/build_cmds.go) feeding the HRD-021 bherald binding; hermetic -race green
HRD-042	task	low	§11.4.31 cherald submanifest verify — Submodule-Dependency-Matrix — Batch C: implemented as cherald §43 verify/check command (cherald/cmd/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only; hermetic -race green
HRD-043	task	high	§11.4.36 pherald install-upstreams — install_upstreams wrapper — Batch C: implemented as gitops command (§11.4.74 FindScript-resolved constitution script) feeding the HRD-023 binding; hermetic temp-repo tests -race green
HRD-044	task	middle	§11.4.37 pherald fetch-guard — pre-edit fetch + rebase enforcement — Batch C: implemented as gitops command (§11.4.74 wrap of git binary); --emit feeds HRD-023 binding; hermetic temp-repo tests -race green
HRD-045	task	high	§11.4.40 rherald gate retest — pre-tag full-suite retest gate — Batch C: implemented as rherald §43 release command (rherald/cmd/rherald/release_cmds.go) feeding the HRD-022 rherald binding; reuses shared commons/gitops; hermetic -race green
HRD-046	task	high	§11.4.41 sherald force-push gate — merge-first + per-session-auth enforcement — Batch C: implemented as sherald §43 command (sherald/cmd/sherald/sysops_cmds.go) feeding the HRD-020 sherald binding; PRE-FLIGHT GATE (detects+classifies+exits-nonzero, never performs the op); hermetic -race green; fix so the CLI runs without a JWT secret
HRD-047	task	middle	§11.4.45 / .56 scherald status-digest — periodic Status.md sweep + regenerate — Batch C straggler: implemented in scherald/cmd/scherald/digest_cmds.go feeding the HRD-025 scherald binding (§11.4.45 status-sweep; detect/report + regenerate; regenerates Status_Summary); hermetic -race green
HRD-048	task	low	§11.4.53 cherald fixed-summary sync — standalone Fixed_Summary regenerate — Batch C: implemented as cherald §43 docs-pipeline command (cherald/cmd/docops_cmds.go) feeding the HRD-019 cherald binding; detect-by-default + regenerate; ships the §107 lazy-verifier fix so the CLI runs without a JWT secret; hermetic -race green
HRD-049	task	middle	§11.4.55 pherald reopen <HRD-NNN> — Issues→Fixed reversal + Reopen — Batch C: implemented as gitops command (the Wave 6.5 inbound Fixed→Issues reversal); --emit feeds HRD-023 binding; hermetic temp-repo tests -race green
HRD-050	task	middle	§11.4.59 cherald readme sync — README doc-link regen + multi-formats — Batch C: implemented as cherald §43 docs-pipeline command (cherald/cmd/docops_cmds.go) feeding the HRD-019 cherald binding; detect-by-default + regenerate; ships the §107 lazy-verifier fix so the CLI runs without a JWT secret; hermetic -race green
HRD-051	task	high	§11.4.60 cherald composite-gate — canonical implementation of CM-107 COMPOSITE-SYNC — Batch C: implemented as cherald §43 verify/check command (cherald/cmd/cherald/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only; hermetic -race green
HRD-052	task	middle	§11.4.65 cherald export — bulk Markdown export wrapper (md/html/pdf) — Batch C: implemented as cherald §43 docs-pipeline command (cherald/cmd/cherald/docops_cmds.go) feeding the HRD-019 cherald binding; detect-by-default + regenerate; ships the §107 lazy-verifier fix so the CLI runs without a JWT secret; hermetic -race green

ID	Type	Criticality	Title
			regenerate; ships the §107 lazy-verifier fix so the CLI runs without a JWT secret green
HRD-053	task	high	§11.4.71 pherald pre-push — fetch + investigate + integrate hook — Batch C: implemented as gitops command (§11.4.74 wrap of git binary); --emit feeds HRD-023 binding; repo tests -race green
HRD-054	task	middle	§11.4.73 cherald spec-version check — Revision-vs-edits drift detected; implemented as cherald §43 verify/check command (cherald/cmd/cherald/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only; hermetic -race green
HRD-055	task	middle	§11.4.74 cherald catalogue-check <pr> — scan PR for Catalogue-Catalogue-Checker runner over vasic-digital + HelixDevelopment — Batch C: implemented as cherald check command (cherald/cmd/cherald/checkops_cmds.go) feeding the HRD-019 cherald binding; detect-only; hermetic -race green
HRD-056	task	high	§12.6 sherald mem-budget watch — daemon-mode 60% threshold watermark; .host.safety_breach — Batch C: implemented as sherald §43 command (cherald/cmd/sherald/sysops_cmds.go) feeding the HRD-020 sherald binding; detect-only; GATE (detects+classifies+exits-nonzero, never performs the op); hermetic -race green; verifier fix so the CLI runs without a JWT secret
HRD-085	task	middle	Implement single-task admin operations on commons_infra.pgxTaskRepository: GetByID, Update, Delete. Required for Requeue + MoveToDeadLetterCache + operator tooling. Currently all three return ErrUnsupported (HRD-085). Scope: BackgroundTask row by ID; full-row Update including JSONB columns; soft-delete decision per §107.
HRD-086	task	middle	Implement worker-side running-task state mutations on commons_infra.pgxTaskRepository: UpdateStatus, UpdateProgress, UpdateHeartbeat, SaveCheckpoint. Required by River workers + the task reaper. Currently all four return ErrUnsupported (HRD-086). Scope: state mutation guarded by valid-state-machine check; progress as (float64, message); last updated_at = now() writeback; checkpoint as opaque []byte blob persisted to task row.
HRD-087	task	low	Implement admin / stats / audit reads on commons_infra.pgxTaskRepository: GetByStatus, GetPendingTasks, CountByStatus, GetTaskHistory. Batch C: v1/queue/* introspection endpoints + Peek delegate. Currently all four return ErrUnsupported (HRD-087). Scope: status-filtered paginated reads; per-task Peek; status-bucketed counts; per-task execution-history rows from the task_execution_history table.
HRD-088	task	middle	Implement queue-recovery reads on commons_infra.pgxTaskRepository: GetStaleTasks, GetByWorkerID. Drives the stuck-task reaper + worker-assignment loop. Currently both return ErrUnsupported (HRD-088). Scope: stale-task lookup against updated_at; per-worker assignment lookup for re-dispatch after a worker dies.
HRD-089	task	low	Implement resource-monitor I/O on commons_infra.pgxTaskRepository: SaveResourceSnapshot, GetResourceSnapshots. Persists per-task resource samples for capacity-planning + post-mortem. Currently both return ErrUnsupported (HRD-089). Scope: append-only insert keyed by (task_id, captured_at); paginated read in reverse-chronological order. Schema target: a task_resource table not yet in the §9.6 migration set — adding it is part of this HRD.
HRD-116	task	middle	Wave 7 T7 — qaherald Slack MessengerClient + channel-keyed builder (cherald/internal/messenger/{slack.go, builder.go}): raw-HTTP client for 11 MessengerClient methods (Me/Send/SendPhoto/SendDocument/SendVideo/SendGif/WaitForReply/Download/Preflight/GetChatMember/Close); GetUpdates via conversations.history+oldest=<ts> watermark (Slack has no getConversations endpoint)

ID	Type	Criticality	Title
HRD-118	task	high	MessageID = dotted-float ts string; channel-keyed Build(BuildConfig) unknown→error). Token NEVER in errors — sanitizeError() defense + TestSlack_Send_ErrorDoesNotLeakToken pins it with a planted xoxb-SUPERSECRET-must-not-leak-7777. 30 hermetic httpstest tests -race time pin var _ MessengerClient = (*SlackClient)(nil). KNO CLEANUP: ships files.upload V1 (deprecated by Slack 2025-Q1 in favor — functionally equivalent for the httpstest seam, swappable without API breakag slack.go header. Wave 7 T9 — e2e_bluff_hunt.sh E124-E131 multi-channel invariants: E124 cha resolution, E125 tgram self-filter post-Wave-7 (pure-refactor invariant), E126 sl (interface satisfied + auth.test + chat.postMessage wire bytes + registry wiring), Slack (env-gated HERALD_SLACK_BOT_TOKEN+HERALD_SLACK_CHANNEL_ID+TestSl triple-guard to prevent silent "no tests to run" PASS-bluff), E128 per-channel in E129 generalized IsSelfEcho (Username+UserID+empty-self), E130 multi-char wiring, E131 qaherald messenger.Build dispatch. sc_anchor on the T6+T7 comr Wave 7 T10 — paired §1.1 mutation gate tests/test_wave7_mutation. PASS / 0 FAIL): M1 blank IsSelfEcho → TestIsSelfEchoUsername hazard caught) → restored → PASS; M2 drop slack channels.Register in TestSlackRegisteredViaInit FAILs → restored → PASS; M3 drop sl append in send.go → TestSlackSendReplyUsesThreadTS FAILs → + quiescence assertion (no MUTATED W7 - residue in 3 production target files Reuses Wave 6 canonical .git/MUTATION_IN_PROGRESS lockfile + trap-c flight dirty-tree-and-residue guard + byte-for-byte assert_restored. FOR ONLY (§107.y / no-concurrent-mutation operator mandate). Wave 7 T11 — §11.4.85 multi-channel stress + chaos. Extended HRD-126's ph pherald/listen_stress_chaos_test.go (NOT duplicated to a new TestListen_MultiChannel_StressBurst (2 channels × 200 concurr ONE runListen — pins no-loss/no-double + per-channel attribution via perChannelCountingReplier keyed on rcpt.Channel proving the c NOT cross-wire; throughput ~9-19k ev/s under -race; clean shutdown <3s + no TestListen_MultiChannel_ChaosOneChannelDies_OtherKeep fires 1 event then errors; slack ticks 50 events at 2ms — pins concurrency-of-f via conjunction: ≥5 slack dispatches landed BEFORE fail-loud teardown AND t surfaces tagged channel="tgram-stub" + subscribe). 12/12 PASS ov -count=3 (3 iterations × 2 new + 2 sibling HRD-126); race-detector clean. ru found: fail-loud-tears-down-siblings (per-channel supervised restart is a deferr doc-comment warns against silent change). Evidence anchors: multichannel_no_drop_under_burst=1 (stress) + multichannel_concurrency_proven=1 + multichannel_one_channel_death_surfaced=1 (chaos). Wave 7 T12 — docs/guides/MESSENGER_CHANNELS.md operator guide sections + 3 siblings .html/.pdf/.docx): §1 architecture (commons.Channel outbo commons_messaging/channels.Channel inbound + registry + IsSelfEcho + per- ASCII fan-in diagram), §2 HERALD_CHANNELS selector (fail-loud on unknow Telegram live setup (BotFather → token → chat-id), §4 Slack live setup (api.sla Socket Mode + scopes + xoxb-/xapp- tokens; §107 token-redaction sanitizat TestSlack_Send_ErrorDoesNotLeakToken), §5 inbox isolation unde {HERALD_INBOX_DIR}/<channel>/<sha256>.<ext>, §6 anti-echo-l (cross-the-wire-then-cache contract), §7 URL schemes for both adapters, §8 14- channel" implementer checklist + companion "what you do NOT touch" list (fra come for free), §9 10-entry troubleshooting cookbook (echo-loop / channel_not
HRD-119	task	high	
HRD-120	task	high	
HRD-121	task	low	

ID	Type	Criticality	Title
			invalid_auth / wrong inbox / HERALD_CHANNELS=unknown), §10 6-checkl deploy audit mirroring OPERATOR_CREDENTIALS.md style, §11 references V3 §11/§32.2/§43 + Wave 7 plan + adapter source paths + every HRD-110..121 REGEN_ALL=1 bash scripts/export_docs.sh (new-guide bootstrap
HRD-148	feature	high	ATMOSphere integration WS-1 — commons_workable SQLite SSoT for (LANDED + tested 2026-05-29). New L1 module: SQLite store (pure-Go mod sqlite, no CGO) mirroring the workable-items schema v2 (items PK (atm_id,current_location) + item_history + meta, WAL, 10-v set); full CRUD repo (Create/GetByID/Update/Delete/List, loud not-found, close validation); per-property Diff(prev,curr) change-feed emitting item.c item.deleted/item.status.changed/item.field.changed/ item.content.updated with exact old→new; ParseTracker matching REAL ## §... [ATM-NNN] + **Status:**/**Type:**/**Severity: real-SQLite tests -race -count=3 green. Keystone for R1/R3/R6.
HRD-149	feature	high	ATMOSphere integration WS-2 — commons_watch file/DB watcher (LANDED + tested 2026-05-29). New L1 module: fsnotify watcher (direct dep) over the .db SSoT debounce-coalesce + SQLite-WAL polling fallback (WAL writes hit the -wal s raise reliable fsnotify Write on the main .db); graceful ctx-cancel, leak-free. 4 -race green. Feeds R1/R2.
HRD-151	feature	high	ATMOSphere integration WS-3 — change→CloudEvent→Runner bridge + Jira per-property-diff message formatter (reuse the existing pherald/internal fan-out; add only the workable-event producer + formatter).
HRD-152	feature	high	ATMOSphere integration WS-4 — inbound action registry + item.update/i investigation.start + ItemMutator (writes commons_workable orchestrator (act-with-confirmation per operator 2026-05-29).
HRD-153	feature	middle	ATMOSphere integration WS-2 — watcher daemon endpoint (pherald wa watch→diff→notify loop; promote to a wherald flavor if it grows).
HRD-154	task	middle	ATMOSphere integration WS-3 — enforce preference/quiet-hours routing (Pre QuietHours types exist but the resolver ignores them).
HRD-090	task	middle	Dead-letter handling on commons_infra.pgxTaskRepository — Move (was the last ErrUnsupported Batch-D stub). Migration 000014_dead_l (.up/.down, FK→background_tasks ON DELETE CASCADE, no tenant/RLS – Tx-atomic move: INSERT a full JSONB task snapshot into dead_letter_t mark background_tasks.status='dead_letter' (soft-mark, NOT preserve the events/snapshots FK children); JSONB passed as string() per t SQLSTATE-22P02 lesson. New 13th OPERATIONAL constitution event class ClassQueueDeadLetter ("queue.dead_letter") + QueueEvent payload + DeadLetter() emit method (governance set §42.2's 12 classes unchanged; c 12→13); wired nil-tolerantly into the repo via newPgxTaskRepositoryWi (boot.go keeps the nil-emitter — the Foundation plane has no governance bus+ emitting there would be an emit-into-the-void §107 bluff; governance-plane sub HRD-147 follow-up). Best-effort emit AFTER the durable commit — surfaced swallowed. Four-layer closure (CONST-048): 3 hermetic unit tests + real-PG (TestRepoMoveToDeadLetter_RoundTripAndEmit, 3 independent si terminal-state + dlq snapshot + real-bus .queue.dead_letter delivery) + chaos (50 concurrent moves race-clean, emit-failure-atomic, cancelled-ctx-no-o §1.1 mutation gate (tests/test_hrd090_mutation_meta.sh, 4/0) + e -race green.
HRD-133	task	high	Tgram-completeness Wave B — token-redaction symmetry (commons_messa channels/tgram/errors.go + send.go wraps). sanitizeTgramError token) string strips the bot token from every Send/SendReply error messa

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HRD-134 task high

can NEVER appear in observable output — 6 error-return sites wrapped (ensured withRetry429 paths in both Send + SendReply).
TestTgram_Send_ErrorDoesNotLeakToken plants 123456:supersecret-token-leak-9999, drives an httpstest 401 path, asserts the token substring is absent from returned err.Error() (mirrors slack's TestSlack_Send_ErrorDoesNotLeakToken). Unit-level TestTgram_SanitizeTgramError_Unit (4 subtests) pins the token to be captured in the source: %w error-wrap replaced with %s + sanitizeTgramError(err.Error(), token) since strings.Repeat operates on the string form — the error chain is sacrificed to guarantee the secret is never leaked. Composes with HRD-134 (the 429 retry's log line + final error are themselves secret, logging never leaks the token).

Tgram-completeness Wave B — Bot API rate-limit 429 + retry_after honored (commons_messaging/channels/tgram/send.go + tgram.go sleep errors.go::extractRetryAfter). withRetry429(ctx, fn) errors telebot bot.Send call sites (Send text body + SendReply text + SendReply payload out). On 429 + retry_after=N → sleep N seconds (ctx.Done()-aware) **attempts** total, clamp per-retry wait to **60s** (defense-in-depth against Bot API spam). retry_after extracted via errors.As(err, &telebot.FloodError{}). RetryAfter(telebot's typed error — NOT Logged retry line goes through sanitizeTgramError so the token stays out (composes with HRD-133). Injectable sleeper field on Adapter for hermetic + fake-clock both supported via SetSleeperForTest). 4 hermetic httpstest tests (HonorsRetryAfter / GivesUpAfter3Attempts / RespectsCtxCancel / ClampsRetries) unit test (ExtractRetryAfter_Unit) — 5 total, all -race -count=1 green.

HRD-135 task middle

Tgram-completeness Wave A — edited-message + bot-kicked typed-error + service ignore-and-log handlers (commons_messaging/channels/tgram/subscriber bot.Handle(telebot.OnEdited, ...) builds the same InboundEventHandler does + Raw["edited"]=true; bot.Handle(telebot.OnMyChatMember botKickedFromUpdate detects status transitions to kicked/left/restricted **for specifically** (Alice-left + promotion-to-Admin false-positive guarded) and emits log.Printf "tgram: bot kicked from chat: chat_id=... new ErrBotKicked"; serviceHandler registered against 15 telebot service Calls (UsersJoined / UserLeft / NewGroupTitle / etc.) emits NOTHING to InboundHandler tgram: ignored service message kind=... chat_id=.... Sentinel ErrBotKicked = errors.New("tgram: bot kicked from chat") (callable via errors.Is). The OnMyChatMember closure returns nil deliberately poller keeps running for OTHER chats. 17/17 PASS -race -count=1 (3 new subscribe_lifecycle_test.go + 14 existing, zero regression).

HRD-136 task low

Tgram-completeness Wave A — Bot API webhook secret_token validation (commons_messaging/channels/tgram/webhook.go). WebhookHandler http.Handler validates X-Telegram-Bot-API-Secret-Token via cryptoutil.Subtle.ConstantTimeCompare (timing-attack-safe); refuses empty secret token (NewWebhookHandler("", ...) errors loud — §107 footgun guard); refuses empty body on header mismatch / missing header (no token-existence-oracle leak 400 on malformed JSON body; 405 on wrong method (POST-only); 500 (no body) InboundHandler error. Forwards a decoded tele.Update as the SAME Inbound shape OnText/OnEdited produce — unified downstream contract. **Skeleton** pherald listen --webhook + the setWebhook(secret_token=...) the public-endpoint TLS listener are deferred to V1.x; the cryptographic-validation tested so the future flip is a one-flag affair. 7/7 hermetic httpstest tests -race -ValidSecret_Dispatches, InvalidSecret_Returns401_NoTokenLeak, MissingSecret

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HRD-137	task	high	GET_Returns405, MalformedJSON_Returns400, EmptySecretAtConstruction_ConstantTimeCompare. Tgram-completeness Wave C — §11.4.85 stress + chaos suite (commons_mes channels/tgram/stress_chaos_test.go). TestTgram_Stress_MultiRecipientFanOut drives the REAL Adapter.SendReply via NewAdapterWithBaseURL httptest seam under iterations = 400 total calls against 20 UNIQUE chat_ids on the SAME *tgram asserts no-loss (every chat_id received exactly 20 hits), no-double (global hit confusion (uniform per-chat distribution, no cross-wire), -race-clean, throughput (observed 5215-6422 ev/s, p99=9.73-15.20ms). TestTgram_Chaos_GetUpdatesPollerResilience converted from with-reason to a real httptest-driven fault-injector PASS via a ONE-LINE subfix: threads a .baseURL into telebot.Settings.URL before NewBot (p ensureBot pattern at tgram.go:60-67; production path unchanged when " "). 8-phase fault matrix (3× 500 + 2× 1.5s hangs honoring ctx-cancel + 1× mic connection-close + 2× success), bot-username-aware self-filter bypass (success from @someuser, not chaostestbot). Deterministic across -race -c §11.4.50 — all 3 iterations IDENTICAL: recovery_event_count=6 fault_modes(500=12 hang=8 close=3 ok=6) subscribe_err deadline exceeded goroutines 3→4 (leaked=1) duration=tgram suite 26/26 PASS, 0 SKIP post-fix (was 24/24 with chaos SKIPPed). §11 captured-evidence per-iteration table; P2 blast-radius — subscribe.go base only when unused (production unaffected); P3 cross-feature — the seam mirror ensureBot pattern, no novel state; P4 deep-research — telebot.v3 LongPoll under fault (poller logs + continues, no internal retry); P5 anti-bluff — handlerF atomics, ctx-aware subscribe return, no goroutine leak (delta=1) all positive cap from REAL Subscribe poller, not synthetic. Tgram-completeness Wave A — docs/guides/TELEGRAM.md per-module dive guide (810 lines, 11 §-sections, 58 subsections, 3 siblings .html/.pdf/.docx) overview (inbound getUpdates + outbound Send/SendReplyGeneric); walkthrough (/newbot → token → /setprivacy); §3 required env table + HERALD_TGRAM_CHAT_ID (3 methods: getUpdates curl / @userinfobot / @R pre-deploy smoke-test commands; §5 attachment size limits (4096 text, 10/50/2 50MB document Bot-API-cap workaround = local Bot API server) + the conten path; §6 chat-type matrix (private/group/supergroup/channel id formats + requir per type); §7 reply threading via reply_to_message_id + the attachment t (Wave 6.5 mutation gate M3); §8 self-filter contract (cross-the-wire getMe + c loop hazard if skipped — Wave 6 T12 M1); §9 12-entry troubleshooting cookbo (401/400/429/413/403/echo-loop/long-poll-stuck/409/privacy/webhook/zero-by §10 6-subsection pre-deploy operator audit (tick-box checklist mirroring OPERATOR_CREDENTIALS.md style); §11 references (spec V3 §11.1/§32.2/ HRDs + source paths). Cross-linked from existing docs/guides/messeng TELEGRAM.md (HRD-011 setup, preserved untouched). Wave 8 Track B — qaherald/internal/mtproto/ MTProto user-accou wiring (gotd/td v0.144.0 vendored as submodules/gotd-td per §11.4.74 c Client interface (Connect / SendMessage / WaitForReply / WhoAmI / C gotd/td's telegram.Client with session-storage at ~/.config/herald mtproto.session (mode 0600, encrypted MTProto session blob). qahera login/whoami/logout Cobra subcommands drive the AuthFlow: app_id + phone-from-env, 2FA password from env, OOB-code via stdin (TTY by default; HERALD_MTPROTO_ALLOW_NON_TTY=1 env override enables delegated-by piped flow for §11.4.98-compliant headless login). Conductor-driven end-to-end
HRD-138	task	low	Tgram-completeness Wave A — docs/guides/TELEGRAM.md per-module dive guide (810 lines, 11 §-sections, 58 subsections, 3 siblings .html/.pdf/.docx) overview (inbound getUpdates + outbound Send/SendReplyGeneric); walkthrough (/newbot → token → /setprivacy); §3 required env table + HERALD_TGRAM_CHAT_ID (3 methods: getUpdates curl / @userinfobot / @R pre-deploy smoke-test commands; §5 attachment size limits (4096 text, 10/50/2 50MB document Bot-API-cap workaround = local Bot API server) + the conten path; §6 chat-type matrix (private/group/supergroup/channel id formats + requir per type); §7 reply threading via reply_to_message_id + the attachment t (Wave 6.5 mutation gate M3); §8 self-filter contract (cross-the-wire getMe + c loop hazard if skipped — Wave 6 T12 M1); §9 12-entry troubleshooting cookbo (401/400/429/413/403/echo-loop/long-poll-stuck/409/privacy/webhook/zero-by §10 6-subsection pre-deploy operator audit (tick-box checklist mirroring OPERATOR_CREDENTIALS.md style); §11 references (spec V3 §11.1/§32.2/ HRDs + source paths). Cross-linked from existing docs/guides/messeng TELEGRAM.md (HRD-011 setup, preserved untouched).
HRD-139	task	high	Wave 8 Track B — qaherald/internal/mtproto/ MTProto user-accou wiring (gotd/td v0.144.0 vendored as submodules/gotd-td per §11.4.74 c Client interface (Connect / SendMessage / WaitForReply / WhoAmI / C gotd/td's telegram.Client with session-storage at ~/.config/herald mtproto.session (mode 0600, encrypted MTProto session blob). qahera login/whoami/logout Cobra subcommands drive the AuthFlow: app_id + phone-from-env, 2FA password from env, OOB-code via stdin (TTY by default; HERALD_MTPROTO_ALLOW_NON_TTY=1 env override enables delegated-by piped flow for §11.4.98-compliant headless login). Conductor-driven end-to-end

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HRD-140	task	high	contribution = one-time paste of 5-digit Telegram OOB code at first-run login (fundamentally requires this; session persists thereafter for unlimited reruns). HR verified leak-free across all evidence (token-shape regex + 32-hex api_hash regex; runtime artifact removed; .gitignore hardened). Wave 8 Track B Test 1 — TestMTPProto_Subscribe_AutonomousRoundtrip (qaherald/internal/lifecycle/mtproto_subscribe_test.go, build-tag integration_mtpproto). §11.4.98 replacement for the legacy TestSubscribe_LiveBotAPI Bot-API-only path which is fundamentally broken (a bot can never send-to-itself, so the legacy test required an operator typing in a stimulus) → poll bot.getUpdates for delivery confirmation → assert message delivered within timeout. Honest-SKIP per §11.4.3 when MTPProto creds absent OR session expired (never falls back to manual-dep path). Conductor-verified PASSING LIVE in 1.4.98. §11.4.98-pure interpretation (chain runs without human action during execution).
HRD-141	task	high	Wave 8 Track B Test 2 — TestMTPProto_Wave6_AutonomousClosedLoop (qaherald/internal/lifecycle/mtproto_wave6_loop_test.go, build-tag integration_mtpproto). §11.4.98 replacement for tests/test_wave6_live_loop.sh (which required manual operator typing). Spawns pherald to listen as a child process with cmd.Dir = repoRoot (CRITICAL — pherald is in docs/Issues.md relative to its CWD), pre-acks pending updates via consumePendingUpdates(t, botToken) helper (calls bot.getUpdates(offset=max+1) to ACK stale chat messages BEFORE pherald starts, preventing TLS-connection-reset storms from prior runs). MTPProto sends stimulus based §11.4.98 autonomy assertion polls pherald's --qa-out-dir JSONL trace for telegram.message(in) + cc.dispatch + cc.reply directions, matched on the full text string rather than message_id (MTPProto's int64 message_id is a different namespace than API's chat-local message_id — §11.4.98 contract is "chain runs end-to-end with human action", not "Claude generates perfect content"). PASSED LIVE in 48.58s.
HRD-142	task	high	Wave 8 Track B Test 3 — TestMTPProto_Wave65_LifecycleAutonomousRoundtrip (qaherald/internal/lifecycle/mtproto_wave65_lifecycle_test.go, build-tag integration_mtpproto). §11.4.98 replacement for the legacy lifecycle - -n interactive mode (which required operator typing per-scenario stimuli). Drives 3 Wave 6.5 fast-path scenarios — Help / Status / Continue — autonomously via MTPProto subtest spawns pherald + sends MTPProto stimulus + asserts on the pherald journal for telegram.message in + `cc.reply`.
HRD-018	task	high	commons_constitution Go package: Evaluator interface + 12 event-class constants + constitution_state/constitution_bindings migrations + bundle-hash + mode-ladder runtime config + constitution_audit write-through (the closing g Evaluators framework, emit helpers, bundle-hash, mode-ladder, state.MemoryAudit, state.Postgres, ladder.Memory/ladder.Postgres, and migration 000006/000007/000008 were all coded in rl.v1.0.0 Batch A closed the remaining bluff: migration 000006 defined constitution_audit but NO code ever wrote to it. Runner.Run set RunOutcome.Audited=true with nothing persisted. Added AuditStore interface (audit.go) + state.MemoryAudit (append-only interface) + state.PostgresAudit (RLS-guarded INSERT-only into constitution_audit; append-only is enforced by the migration's no-update/no-delete RLS policies). Pherald now writes a durable audit row for every CHANGED transition in ModeWarn/ModeEnforce — the EXACT emitted event ID (new IDemitter optional interface) + PolicyViolationID/PolicyClearedID; emit returns the generated event ID. NewRunner gained a required AuditStore arg (nil = hard error). §11.4.92 multi-j

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HRD-026	task	middle	RED (compile-fail before backends) → GREEN, -race -count=3 determining DB-level SELECT shows enforce rows carry emitted_event_id, warn rows NULL Constitution-bundle hash captureer — computes SHA-256 of rendered Constitution evaluation time; persists on every emitted event for replayability. The captureer (Capture/CaptureBytes/Captureer) was coded in rl and the hash already emitted-event metadata (bundle_hash hex). v1.0.0 Batch A completed the per 32-byte bundle_hash is now stored on every constitution_state row and constitution_audit row (verified by the DB-level SELECT ... encode(bundle_hash, 'hex') in the evidence dir showing a 64-hex SHA256 audit row), closing the §42.1.3 replay-correlation requirement. Mode-ladder runtime config (constitution_bindings table + admin REST surface flip allow/warn/enforce per binding per tenant without redeploy). The ModeLadder.Memory/ladder.Postgres (reading constitution_bindings table evaluation time) were coded in rl. v1.0.0 Batch A added the admin REST surface: cherald.modes — GET /v1/compliance/modes (list bindings + unbound-defaults) and compliance/modes/:rule_id (one binding's effective mode, enforce default) and compliance/modes/:rule_id (flip allow/warn/enforce; records the operation mutated_by). JWT-gated via the existing commons_auth.GinMiddleware. Mounted in cherald serve (a single shared PG pool backs both compliance + cherald.bluff: 401 no-auth / 401 bad-tenant / 400 bad-mode (loud, never a silent default), proof: a PUT flip to warn is reflected by a subsequent GET AND by ladder .C evaluation hot path — runtime config, no redeploy.
HRD-027	task	middle	§11.4.85/§32.1 Runner idempotency-window FIX — concurrent same-key replay duplicate-dispatches a notification (FINDING from HRD-125 stress/chaos). Rollback idempotency.go: 64-72 treated a Redis-SETNX-miss-with-no-PG-row as a success (§32.1 "Redis-lies-PG-truths" design) while the events_processed archive was written at Stage 7 (OutcomeRecorder.Process) — so concurrent same-key dispatch arriving in the Stage-2→Stage-7 window ALL dispatched (observed 2–4 of a 1000 archival exactly-once via PG UNIQUE + ON CONFLICT DO NOTHING, but only bounded = duplicate-alert delivery under load). Fix: claim the idempotency BEFORE dispatch — INSERT events_processed ... ON CONFLICT DO NOTHING moved to Stage 2 (the PG row, serialized by PRIMARY KEY(tenant_id, idempotency) now the AUTHORITATIVE exactly-once dispatch gate; 0-rows-affected = duplicate dispatch), preserving the nil-Redis PG-only degrade. ALSO closed the latent run CachedRcpt.WasReplay data race (mutated a possibly-shared *Receipt). HRD-125 stress test was upgraded to assert sends==1 under the same 1000× stress — the stronger assertion HRD-125 deliberately could NOT make pre-fix (§11.4.85 on-discovery). §11.4.92 multi-pass: Pass-1 independent -race -count=3 run storage green; Pass-5 sends==1 is code-true (RED proved before fix). wave3 retired (4f45f26) — Stage-7 Insert now redundant, M4's blast-radius moved to claim.
HRD-132	bug	high	§11.4.85 / §1.1 paired mutation gate for the GAP-3 stress+chaos suite — test test_stress_chaos_mutation_meta.sh (GAP-3 plan §5 / §6 T9). Proving a hermetic stress/chaos detector is genuinely LOAD-BEARING via the canonical test_wave6.5_mutation_meta.sh idiom: .git/MUTATION_IN_PROGRESS + dirty-tree + pre-existing-residue refusal pre-flights, run_paired (apply perl marker landed → run detector EXPECT-FAIL → git checkout restore → b2 assert_restored vs HEAD → re-run EXPECT-PASS), and a final check scan asserting no MUTATED SC-M* markers leaked (§107.y). FIVE paired mutations against a hermetic detector: SC-M1 drop ctx from claude_code dispatch.go exec.CommandContext→exec.Command → timeout-cancel test no longer sleep → watchdog FAIL; SC-M2 swallow the claude_code parseReply deco
HRD-130	task	high	§11.4.85 / §1.1 paired mutation gate for the GAP-3 stress+chaos suite — test test_stress_chaos_mutation_meta.sh (GAP-3 plan §5 / §6 T9). Proving a hermetic stress/chaos detector is genuinely LOAD-BEARING via the canonical test_wave6.5_mutation_meta.sh idiom: .git/MUTATION_IN_PROGRESS + dirty-tree + pre-existing-residue refusal pre-flights, run_paired (apply perl marker landed → run detector EXPECT-FAIL → git checkout restore → b2 assert_restored vs HEAD → re-run EXPECT-PASS), and a final check scan asserting no MUTATED SC-M* markers leaked (§107.y). FIVE paired mutations against a hermetic detector: SC-M1 drop ctx from claude_code dispatch.go exec.CommandContext→exec.Command → timeout-cancel test no longer sleep → watchdog FAIL; SC-M2 swallow the claude_code parseReply deco

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HRD-129	task	high	truncated-reply test returns nil → FAIL; SC-M3 make inbound extractReply default: panic instead of return ...err → degrade test panics → FAIL; events mapErrorToStatus event_parser: 400→500 → input-corruption FAIL; SC-M5 swallow the commons_storage applyMigration disk-full write ENOSPC reports success → silent-swallow guard FAIL. Runs ONE-AT-A-TIME FOREGROUND (§107.y + the 2026-05-27 concurrent-gate-residue memo). Co result: 6 PASS / 0 FAIL (5 mutations load-bearing). §11.4.85 stress+chaos suite wired into the canonical anti-bluff harness script e2e_bluff_hunt.sh as new invariants E81-E88 (GAP-3 plan §4 / §6 T8) — ONLY append after E80 (E1-E80 logic untouched). Each invariant RUNS a RE stress/chaos test via the existing check helper AND cites the captured-evidence docs/qa/<run-id>/stress_chaos/ as its §11.4.2/§11.4.5 positive-evidence through a new sc_anchor helper that greps a SPECIFIC load-bearing value (artefact FAILs as a §11.4 PASS-bluff; absent evidence dir SKIPs-with-reason). runner exactly-once-archive + bounded dispatch (HRD-125); E82 /v1/event + duplicate-key + auth-storm (HRD-123, anchor events/categorised_event all_malformed_rejected_no_5xx); E83 /v1/compliance fail-loud /v1/safety_state concurrent-mutation (HRD-124); E85 pherald list malformed-Raw panic-free + extract-degrade (HRD-126); E86 claude_code pro timeout-cancel + truncated-reply (HRD-127); E87 disk-full tagged-error ENOS E88 §12.6 host-mem headroom (HRD-128). All eight hermetic; live container-p fill / container-OOM variants are SKIP-with-reason inside the Go tests (HERALD_STRESS_LIVE_*=1 gated). Verified bash -n clean; E81-E88 block deterministic; header bumped E1..E80→E1..E88. Full scripts/e2e_bluff reports 78 PASS / 0 FAIL (was 60; +18).
HRD-128	task	high	§11.4.85 container-orchestrated / resource-exhaustion stress+chaos tests (GAP-3 §6 T7 + §7 host-safety) — commons_storage/resource_stress_chaos (hermetic core) + tests/test_resource_stress_chaos.sh (LIVE core HERMETIC PASS (×2 deterministic): (1) disk-full error propagation — a fake database.Database.Tx injects syscall.ENOSPC on the migration-UP REAL RunMigrations → applyMigration write path; asserts real write error NOT swallowed, error tagged commons_storage: apply + exec u errors.Is(err, syscall.ENOSPC) true, 0 migrations reported applied mem headroom — HostMemHeadroom() pre/post a 25× workload shows used_fraction_delta ≈ 0 (host needle did not move) = §12.6 compliance (deterministic SKIP without env); real bounded-scratch disk fill HARD-capped HERALD_STRESS_LIVE_DISK=1. LIVE container scenarios SKIP-with-reason safety-gated); container OOM-confinement + connection-churn against a --memory PG, gated HERALD_STRESS_LIVE_OOM=1 + a §12.6 pre-flight host-headroom REFUSES to add pressure when host used-fraction ≥ 60% (host was ~74% this SKIP). §12/§12.6 honoured: NO host-OOM, NO real-host-disk fill, NO host pro test -race -count=3 green (PASS×3/PASS×3/SKIP×3); go vet clean.
HRD-127	task	high	§11.4.85 claude_code dispatch stress+chaos tests (GAP-3 plan §1 row 5 / §6 T6 commons_messaging/dispatch/claude_code/dispatch_stress_chaos_test.go (in-package, exercises unexported bootstrapSession/parseReply) + 5 committed hermetic shims testd claude-*.sh. Per §11.4.27 only the EXTERNAL claude CLI is faked via New(binaryPath,...) seam; NO production source edited, NO new dep, NO invoked. STRESS: (1) N=24×25=600 concurrent Dispatch (--resume path every reply really parsed, every call resumed the SAME pre-seeded UUID, and p50=62.9ms/p95=96.4ms; (2) N=16 concurrent COLD-START bootstrap — all one anchor UUID persists, -race clean. CHAOS: (a) process-death exit 13

ID	Type	Criticality	Title
HRD-126	task	high	-KILL \$\$ → tagged error, no hang (watchdog), partial stdout NOT accepted; 30s-sleep shim vs 800ms ctx → cancellation fires, Dispatch returns ~803ms not <<<HERALD-REPLY>>> → explicit decode reply JSON error, never sil FINDING-1: cold-start concurrent bootstrap is last-writer-wins by design (hone not bootstrap_count==1 bluff). FINDING-2: timeout-cancellation latency claude being a single process (hardening direction = process-group kill, flagg owner). go test -race -count=3 deterministically green ×3. §11.4.85 pherald listen inbound-path stress+chaos tests (GAP-3 plan §1 pherald/internal/inbound/stress_chaos_test.go (in-package unexported extractReplyToMessageID) + pherald/cmd/pherald/listen_stress_chaos_test.go (drives the REAL runListen loop). the channel Subscriber + CC CodeDispatcher boundaries are faked; the Dis routing switch + runListen fan-in + clean-shutdown run unmodified. STRES N=16×50=800 concurrent Dispatcher.Handle — 0 errors, exactly-once re (800/800), 0 goroutine leak, p50=0.41ms ~25k dispatch/s; (2) 400-event burst th runListen — nothing dropped, clean shutdown on cancel (<3s), no leak. CH Raw payloads (nil map, missing/nil/string/bool/slice/map message_id) → extractReplyToMessageID degrades to (0,err); valid int/int64/int32/float 64-dispatch corrupt-Raw flood panic-free (recover-guard), degraded replyTo subprocess fault (killed/truncated/marker-less/empty/unknown-action) → stage inbound: error, no reply leak, no hang; (c) once-only side-effect — 1000 red same issue. open collapse to exactly-ONE durable sink row. go test -r deterministically green ×3 on both packages. NO production source edited, NO §11.4.85 Runner 7-stage pipeline stress+chaos tests (GAP-3 plan §1 row 3 / §6 pherald/internal/runner/stress_chaos_test.go exercising the runner.Run (only the external boundary — PG events_processed store, chan SETNX seam — faked per §11.4.27; orchestrator + 7 stages run unmodified). S goroutines × 40 iters concurrent same-key replay + fresh keys, -race clean. C duplicate flood @ 50 parallel under live atomic Redis SETNX; (a') same under fallback degrade; (b) hermetic PG-deadlock (SQLSTATE 40P01) injection on th events_processed Insert. SURFACED HRD-132: dispatch exactly-once was NO under concurrent replay (events_processed archived at Stage 7; observed 2–4 du of a 1000× flood); tests asserted the honest code-true contract (archival exactly-dispatch) pre-fix, then upgraded to sends==1 once HRD-132 landed the Stage test -race -count=1 green. §11.4.85 Gin /v1/compliance (cherald) + /v1/safety_state (sherald stress+chaos tests (GAP-3 plan §1 row 2 / §6 T3) — cherald/internal/c compliance_stress_chaos_test.go + sherald/internal/safe safety_stress_chaos_test.go, each driving the REAL flavor handler process httptest server dialed by net/http.Client through the product (cli.TOONMiddleware → auth → handler). Only the external boundary fal NO production source edited, NO new dep. STRESS (each flavor): N=10×200= -race clean — 0 errors, 0 5xx, all 200 (compliance ~8.4k req/s, safety_state ~ Content-negotiation correctness under concurrency — every request alternates / application/toon vs application/json → 100% Content-Type mat Accept header (0 codec-bleed), every toon body real TOON wire bytes. CHAOS PG-drop (errStore) → 2000/2000 fail-loud 500 each naming the failed depe fabricated 2xx (a swallowed-error 200 {results: []} would falsely tell an o violations" — the §107 mode this gate forbids); safety_state has no hot-path DE code-true analogue (concurrent state-mutation under load, -race-clean RWMu instead; (b) auth storm — 1000 random tokens through the REAL
HRD-125	task	high	
HRD-124	task	high	

ID	Type	Criticality	Title
HRD-123	task	high	commons_auth.GinMiddleware+HMAC(HS256) → 1000×401, 0 bypass; test -race -count=3 green×3. §11.4.85 Gin /v1/events stress+chaos tests (GAP-3 plan §1 row 1 / §6 T2) - internal/http/events_stress_chaos_test.go driving the REAL events handler (EventsHandler → runner.Run, all 7 stages) over a REAL httptest server dialed by net/http.Client through the production Gin (cli.T00NMiddleware + auth gate). Only the external boundary (PG/Redis runner.NewFakeRunner) faked per §11.4.27. STRESS: N=12×200=2400 POSTs, -race clean — 0 errors, 0 5xx, all 202; p50=0.706ms ~13,322 req/s. C duplicate-key-under-load (1200 POSTs, per-worker keys) → graceful degrade, C input-corruption (8 MiB random, truncated JSON, non-UTF8, JSON array, miss empty) × 5 → every body a tagged event_parser: 4xx, 0 panic, 0 5xx; (d) 1000 random tokens through the REAL commons_auth.GinMiddleware → 1000×401, 0 bypass; (a) live PG-drop → SKIP-with-reason (proven hermetic FINDING (→ HRD-132 at HTTP layer): a SHARED-key cross-worker concurr triggers the REAL runner.go:132 CachedRcpt.WasReplay data race (HRD-132). E82 anchor re-captured as all_malformed_rejected_no_5 transport-rejection-of-8-MiB-body is valid). §11.4.85 stress+chaos shared scaffold — Go-native commons/stresschaos package (GAP-3 plan §2 / §6 T1). Provides: a bounded concurrent load-runner (workers × M iterations of a closure via stdlib sync.WaitGroup, no new dep rank percentile math (Percentiles → {p50, p95, p99, max, min, count} latency.json + latency_histogram.csv; an evidence-dir writer creating <run-id>/stress_chaos/<surface>/; and a best-effort host-memory (HostMemHeadroom — vm_stat+sysctl hw.memsize on darwin, /p on linux) for the §12.6 60%-ceiling proof (degrades to probe-unavailable failure). Self-tested: 8 unit tests (percentile correctness on a known 1..100ms data creation, bounded-goroutine proof, host-mem best-effort, §12.6 ceiling predicate test -race -count=1. Lowest sensible layer per CLAUDE.md layering (importable by every flavor).
HRD-122	task	middle	Wave 7 T5 — multi-channel pherald listen (HERALD_CHANNELS far goroutines → 1 dispatcher). pherald/internal/inbound/dispatcher Wave-6 TgramReplier interface (Telegram-native int64 chatID/int r signature) → generic inbound.Replier with SendReply(ctx, recip commons.Recipient, body string, replyToID string, att []commons.Attachment) (string, error) — recipient + string ids satisfy it. Config.TgramReply → Config.Reply; NewDispatcher n Dispatcher.reply field + the Handle "reply" branch + both fastPath migrated to build a commons.Recipient{Channel, ChannelUserID} strconv.Itoa(replyToID) (" " when 0). pherald/cmd/pherald/T listenConfig.Subscriber(singular) → Subscribers map[string InboundHandler) error; listenConfig.Replier inbound.Tg inbound.Replier; new loadEnabledChannels() parses HERALD_C (comma-split, trim, dedupe; unset/empty → ["tgram"], Wave-6 single-chann preserved); new perChannelConfig(name) reads namespaced env (HERALD_TGRAM_BOT_TOKEN/HERALD_TGRAM_CHAT_ID, HERALD_SLACK_BOT_TOKEN/_APP_TOKEN/_CHANNEL_ID) → channe loud on missing required creds); loadListenConfigFromEnv resolves ea via the channels.New(name, cfg) registry (T2), builds Subscribers[name]=ch.Subscribe + a channelReplier{ch} shi channels.Channel.SendReplyGeneric → inbound.Replier.Se two intentionally-differently-named generic methods bridged) per channel, and
HRD-114	task	high	

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Type Criticality Title

HRD-113 task high

channelRouter that dispatches each reply to the adapter for recipient. loud no replier for channel %q on unknown — never silently dropped fans in one goroutine per subscriber (manual sync.WaitGroup + buffered err new dep — errgroup-equivalent semantics) all feeding ONE dispatcher; the first subscriber error cancels siblings + surfaces (T11 chaos fail-loud). tgram is blank listen.go for its init() registry registration. journal.go journalingRe to wrap inbound.Replier (records channel+chat_id+reply_to_me strings; JSONL kind kept tgram.send_reply for transcript back-compat). resolution (Wave 7 plan Step 4): inbound.Replier.SendReply keeps the name (the inbound package's own, independent of channels.Channel.SendReplyGeneric); production wiring bridges v channelReplier shim so the journal decorator + dispatcher both stay channel evidence: new TestRunListenMultiChannelFanIn (2 stub subscribers publish 1 event) asserts BOTH messages dispatched AND both replies recorded (fake)] >= 2) — NOT "both goroutines started"; mutation-proven load-b the fan-in loop → seen == map[] → FAIL). Existing single-channel TestListenWiresHandlerAndExitsOnCancel + TestRunListenRejectsBadConfig migrated to the one-entry Subscr PASS. recordingReplier/stubReplierJournal stubs (dispatcher_test journal_test) migrated to the generic signature; numeric chat_id/reply assertions decode.go build ./pherald/... + go test -race -count=1 . pherald/... ./commons_messaging/... all green; go vet clean; clean.

Wave 7 T4 — generalize the bot self-filter via BotSelfIdentity (channel-a commons_messaging/channels/selffilter.go adds the Raw-map (RawSenderIsBot/RawSenderIdentityKind/RawSenderIdentity) StampSender(raw map[string]any, isBot bool, kind Identity value string) (records the sender's native identity into ev.Raw; nil-map IsSelfEcho(ev commons.InboundEvent, self SelfIdentity §32.9 anti-echo-loop guarantee made channel-agnostic — compares the right na IdentityKind: username for Telegram, user_id for Slack, address f deliberately narrow: a DIFFERENT bot is KEPT (multi-bot collaboration is rea — the §107 anchor that the filter is not over-broad); a human is KEPT; an empty never echoes (conservative KEEP → surfaces as duplicate traffic, not silent loss Subscribe refusing to boot on empty self). commons_messaging/channel subscribe.go migrated: captures self := channels.SelfIdentity channels.IdentityUsername, Value: bot.Me.Username} once (keeping the existing empty-username boot-refusal echo-loop guard) and routes handlers (OnText/OnPhoto/OnDocument/OnVoice) through a new stampAndIsSelfEcho(ev, msg, self) helper that stamps the Telegra IsBot+@username into ev.Raw and delegates to channels.IsSelfEcho the four Wave-6 shouldDropBotSelf(msg, selfUsername) early-ret native shouldDropBotSelf func + its TestSubscribeBotSelfFilter SelfFilterForTest (export_test.go) are PRESERVED BYTE-IDENTICA paired-mutation gate (tests/test_wave6_mutation_meta.sh) anchor perl -i -0pe regex on shouldDropBotSelf's body, so keeping it byte-M1 load-bearing (verified: the M1 regex still matches + applies its W6-M1 mar migration). The live path and the gate now express the SAME §32.9 invariant — the generalized IsSelfEcho, the gate on the tgram-native shape. §107 eviden TestIsSelfEchoUsername proves all three username cases — bot-own D different-bot KEPT, human KEPT; TestIsSelfEchoUserID proves the Sla (matching bot id DROPPED, different id KEPT);

ID	Type	Criticality Title
HRD-112	task	middle TestIsSelfEchoEmptySelfNeverEchoes proves empty-self KEEP; the builds events via the production StampSender (DRY — the test can never drop key contract); TestSubscribeBotSelfFilter still PASSES (Wave-6 sha test -race -count=1 ./commons_messaging/... ./pherald go vet clean. Live runtime evidence (real Telegram getMe self-identity + own the closed loop) lands via Wave 7 T9 e2e invariant. Wave 7 T3 — per-channel inbox subdirs ~/ .herald/inbox/<channel>/<sha256>.<ext>. New file commons_messaging/channels/inbox channel-agnostic InboxDir(channel string) (string, error) (r ~/ .herald/inbox/<channel>/, MkdirAll 0700, empty-channel error) + WriteContentAddressed(channel, mime string, r io.Read (path, sha256Hex string, err error) (streams r into <dir>/<sha256>.<ext> via io.MultiWriter(tmp, sha256.New()) — n full payload — then atomic temp+rename; idempotent: if the content-addressed the .part temp is dropped and the existing path returned unchanged; closes r) MimeToExt(mime string) (the Wave-6 tgram-private mimeToExt switch VERBATIM to package level so every adapter shares one map). commons_me channels/tgram/attachments.go DownloadAttachment now does Telegram-specific bot.File(&telebot.File{FileID}) fetch then delete hash-rename to channels.WriteContentAddressed(string(common.Channel mime, rc) — so tgram attachments land under ~/ .herald/inbox/tgram channel subdir, was the flat ~/ .herald/inbox/ in Wave 6). The tgram-private + the crypto/sha256/encoding/hex/io/os/path/filepath imports from attachments.go (the helper owns them now); the method-form (*Adapter).DownloadAttachment(T1) delegates to the package func u forms route through the shared helper. §107 evidence: TestInboxDirIsPer InboxDir("tgram") ends in .../inbox/tgram AND InboxDir("t InboxDir("slack") (no flat-inbox regression); TestWriteContentAddressedHashesAndIsIdempotent pins all the classes on real on-disk state — (a) path shape under inbox/slack/ named < re-read on-disk bytes byte-equal the payload (sha in path == sha of content), (c) content returns same (path, sum) and leaves zero .part residue (real os.R the existing Wave-6 TestDownloadAttachmentContentAddressed (u tgram/ path segment) still PASSES end-to-end against a httptest Bot API routing change is behaviour-preserving. go test -race -count=1 ./ commons_messaging/... ./pherald/... ./qaherald/... all g assumption regressions); go vet clean. Live runtime evidence (real Telegram landing under inbox/tgram/) lands via Wave 7 T9 e2e invariant E83. Wave 7 T2 — channel registry (name→constructor) + init()-based registrati commons_messaging/channels/registry.go adds Register(name Constructor)/New(name, Config) (Channel, error)/Names the ErrUnknownChannel sentinel, the channel-agnostic Config struct (Ch AppToken/Target/BaseURL/Extra), and the Constructor func (Co (Channel, error) type — a sync.RWMutex-guarded map[string]C Register panics on duplicate (a typo in two adapters' init() would silently §107 bluff class this framework forbids; mirrors the qaherald scenario.Reg New wraps ErrUnknownChannel (fmt.Errorf("%w: ...")) for unkn callers (pherald listen) MUST surface it; a silent no-op channel is a §107 commons_messaging/channels/tgram/tgram.go gains an init() channels.Register(string(common.ChannelTelegram), ... constructor builds NewAdapterWithBaseURL(cfg.Token, cfg.Targ
HRD-111	task	high Wave 7 T2 — channel registry (name→constructor) + init()-based registrati commons_messaging/channels/registry.go adds Register(name Constructor)/New(name, Config) (Channel, error)/Names the ErrUnknownChannel sentinel, the channel-agnostic Config struct (Ch AppToken/Target/BaseURL/Extra), and the Constructor func (Co (Channel, error) type — a sync.RWMutex-guarded map[string]C Register panics on duplicate (a typo in two adapters' init() would silently §107 bluff class this framework forbids; mirrors the qaherald scenario.Reg New wraps ErrUnknownChannel (fmt.Errorf("%w: ...")) for unkn callers (pherald listen) MUST surface it; a silent no-op channel is a §107 commons_messaging/channels/tgram/tgram.go gains an init() channels.Register(string(common.ChannelTelegram), ... constructor builds NewAdapterWithBaseURL(cfg.Token, cfg.Targ

ID

Type Criticality Title

HRD-110 task high

HRD-101 task high

cfg.BaseURL) when cfg.BaseURL != "" (http test seam) else
NewWithCreds(cfg.Token, cfg.Target) — so a blank import self-r
adapter. §107 evidence: TestRegistryResolvesRegisteredChannel
actually constructs + returns a working Channel (c.Name()=="fake-rt"
TestRegistryUnknownChannelErrors proves errors.Is(err,
ErrUnknownChannel) for unknown names; TestRegistryDuplicate
the duplicate-Register panic; TestTgramRegisteredViaInit (blank
tgram") proves the init() actually registered tgram in Names() (NOT a
failing before the init() landed: Names()=[dup-rt fake-rt] without
after). The shared fakeChannel test stub uses SendReplyGeneric (the T
interface method name, NOT SendReply). go test -race ./
commons_messaging/... green (tgram regression-free); all 16 workspace
go vet ./commons_messaging/... clean. Live runtime evidence (real
by name + round-trip) lands via Wave 7 T9 e2e invariants E81/E82.
Wave 7 T1 — extract commons_messaging/channels.Channel interface
New package commons_messaging/channels/ defines Channel (embed
commons.Channel — Name/Capabilities/Send/Subscribe/HealthCheck — p
inbound-runtime methods Telegram implicitly defined: SendReplyGeneric
BotSelfIdentity returning SelfIdentity{Kind, Value}, Download
the SelfIdentity value type, and the IdentityKind constants (Identit
IdentityUserID/IdentityAddress). *tgram.Adapter gains BotS
(getMe via ensureBot; empty-username → §107 echo-loop-hazard error), Send
(string→int64/int adapter over the native Wave-6 SendReply), and a method-
DownloadAttachment (routes the package-level func through ensureBot's l
_ channels.Channel = (*tgram.Adapter)(nil) compiles. Naming
from plan Step 4 text resolved + documented in channel.go package doc: th
method is named SendReplyGeneric (not SendReply) so tgram satisfies
touching its native, §107-mutation-gate-anchored int64/int SendReply (whose
generic signature is T5's scope) — keeping T1 a pure refactor with zero cross-p
§107 evidence: var _ channels.Channel = (*tgram.Adapter)(n
assertion in channel_test.go (a renamed method / drifted signature breaks
TestTgramSatisfiesChannel + TestSelfIdentityType PASS; fu
-race ./commons_messaging/... green (tgram regression-free — pur
all 16 workspace modules build; go vet ./commons_messaging/... c
evidence (real getMe self-identity + reply round-trip) lands via Wave 7 T9 e2e i
Wave 6.5 ticket-lifecycle layer — §32.6 deterministic classifier (Bug:/Issue:
Implementation:/Investigation:/Query:/Question:/Help:/St
Continue:/Done:/Resolve:/Reopen:/Override: prefix routing with
Type+Criticality+Confidence scores; natural-language no-prefix → c
DocsIssueOpener (§32.9 — real docs/Issues.md HRD-NNN allocator, con
lockfile) + RunnerEventEmitter (§33.7 — direct runner.Run → CloudEvent
HTTP loopback) + Help:/Status:/Continue: fast-path handlers (bypass
second latency — S2 proved 0.74s, no CC bootstrap overhead) + Done:/Reop
Issues? Fixed migration (rollback-via-sha256 test) + tgram.SendReply ou
fan-out (multipart dispatch) + wiring all of these into pherald listen. §10
(docs/qa/HRD-101-lifecycle-2026-05-23T03-16-17-w6.51 liv
language query — real Opus CC dispatch ~31s incl. bootstrap, threaded reply (v
inbound message_id=30 → reply sent_message_id=31 with
reply_to_message_id=30); S2 Help: fast-path — classifier Type:he
Confidence: 1, CC dispatch ABSENT (event absence + 0.74s latency prove
BuiltinHelp catalogue text returned (reply_to_message_id=32 →
sent_message_id=33); transcript.jsonl captures both directions of all four

ID	Type	Criticality	Title
HRD-100	task	high	query, Help:, Status: also exercised live). Reproducible test harness <code>tests/test_wave6.5_lifecycle.sh</code> covers 15 scenarios (Bug:/Task:/Query: + Done:/Reopen: + classifier edge cases). Spec V3 r13: §32.6/§32.9/§33.6/§33.7 s Tag v0.5.0. Wave 6 — pherald inbound runtime + Claude Code headless bridge (code-doc Cobra subcommand <code>pherald listen</code> runs the closed-loop runtime: Telegram long-poll (OnText/OnPhoto/OnDocument/OnVoice) + bot self-filter (bot anti-echo guard captured at Subscribe construction; refuses to start if <code>getMe</code> returns username) + sha256 content-addressed attachment download into <code>~/herald/<sha256>.<ext></code> (idempotent — duplicate downloads do not re-write) + Claude dispatch with Opus pinned in the spawned argv (<code>claude --model claude resume <UUID> --print <envelope></code>) + envelope pre-text — verbatim wording ("We have received new message from our communication channel <channel-name>. <classification sentence>. <action list>") preceding the <<<HERALD-DISPATCH-v1>>> structured block + <REPLY>>> parser with action routing (reply default / issue.open / ev tgram.Adapter.SendReply(ctx, chatID, text, replyToID, attachments) wiring <code>reply_to_message_id</code> from the original message out-dir <path> flag journals every inbound + outbound event as JSONL p 12 atomic Wave-6 commits T1=ad87d7f (HERALD_PROJECT_NAME resolved + .env.example), T2=0d55274 (Opus model pin), T3=14626f7 (envelope pre T4=b4018a3 (bot self-filter), T5=d9156f1 (OnPhoto/OnDocument/OnVoice download), T6=a22a054 (pherald listen Cobra subcommand), T7=a3b (pherald/internal/inbound/ Dispatcher + action routing), T8=5f92f (tgram.SendReply with <code>reply_to_message_id</code>), T9=f7913b8 (live script tests/test_wave6_live_loop.sh), T10a=520e0c6 (--qa-on JSONL journaling middleware), T11=01bfd1a (e2e bluff_hunt E63-E70 invariant T12=96c7c6b (Wave 6 mutation gate tests/test_wave6_mutation_m paired hermetic mutations). §107 evidence: 8 new e2e invariants E63-E70 land currently SKIP with documented reasons — E63 lifecycle / E64 bot self-filter / sha256 / E66 envelope pre-text / E67 Opus argv / E68 <code>reply_to_message_</code> E69 action routing issue.open / E70 live closed-loop). Conversion SKIP → PAS when the operator exports <code>HERALD_TGRAM_BOT_TOKEN</code> + <code>HERALD_TGRAM</code> <code>HERALD_CLAUDE_BIN</code> and runs <code>bash tests/test_wave6_live_loc out-dir docs/qa/HRD-100-<run-id>/. Tag v0.4.0</code> is DEFERRED T10b live evidence per §107.x) — code-doc closure ships here. pherald /v1/events Runner wiring (REST API surface via Gin Gonic per sp stage event-ingest pipeline lands as 7 concrete structs under <code>pherald/internal</code> (EventParser, IdempotencyChecker, TenantResolver, PolicyGate, SubscriberRes ChannelDispatcher, OutcomeRecorder) orchestrated by <code>runner.Runner.Run</code> <code>internal/http/events.go</code> exposes the live HTTP handler; <code>pherald/{main,stubs}.go</code> lazily build the Runner + JWT verifier inside the serve s other paths still work without <code>HERALD_PG_DSN/HERALD_AUTH_MODE</code>. HTTP Accepted + Receipt JSON on fresh event, 200 + X-Herald-Replay: true missing/invalid JWT, 400 on parse failure. §107 evidence: 23/23 runner package stage + 3 integration scenarios: happy/duplicate/deny); 6 new e2e invariants E3 gate-skipped when absent — honest SKIP-with-reason per §11.4.3); mutation g test_wave3_mutation_meta.sh gains M2 (Duplicate=false unconditional M4 (skip events_processed.Insert → E38/E42 break). M3 SKIP-with-reason per deny-evaluator e2e. Commits T1..T10: eb9b2f4, 40ad60f, 2fdb7b5, 4c4 5468897, 6ea29a4, be71db1, c7b89a4, (this commit).
HRD-016	task	middle	
HRD-011	task	middle	

ID	Type	Criticality	Title
			Telegram channel adapter live integration — telebot.v3 vendored (submodule HealthCheck + Send + Subscribe + outbound_delivery_evidence persistence all §107 bluff guards. Closed atomically with operator-session live evidence 2026-delivery message_id=5 from @atmosphere_worker_bot to chat 2057 (validated via getChat) using the extended wizard CLI's --bot-token (env HERALD_TGRAM_BOT_TOKEN) + --chat-id (validated via getChat) + --non-interactive flags. Token→getMe→getChat→sendMessage chain end in commit 140a2f1 session.
HRD-099	task	low	commons_auth/ JWT verifier — HMAC + JWKS hybrid + Gin middleware + c vendored Herald-internal per §11.4.74 catalogue-check no-match. Catalogue ev catalogue-checks/HRD-099-commons-auth.md. Renumbered from 3a executing-time anchor) in the post-Wave-3a doc-cleanup pass; Wave 3a com dbbea95..0cc6fad keep the old HRD-093 in their commit messages as his
HRD-098	task	middle	sherald /v1/safety_state live — process-local Aggregator + background tick) + Handler with JWT gate via commons_auth.GinMiddleware. Replaces the §107 evidence: e2e E46 (no-auth → 401), E47 (valid HMAC → 200 + current_ last_destructive_op=null + uptime_seconds>=1 + open_events=0). Paired mutat mem Aggregator) proves E47 catches the regression. E48 SKIP-with-reason — trigger awaits HRD-033 body.
HRD-028	task	low	cherald /v1/compliance live — paginated + filter-aware constitution surface with JWT gate via commons_auth.GinMiddleware. Backed by ConstitutionStore.ListQuery extended with Since/Until/Offset (commit 627643 e2e E43 (no-auth → 401), E44 (valid HMAC + empty tenant → 200 + total=0 + page_size=50). E45 SKIP-with-reason — cross-binary cherald-reads-pherald-w Wave 3b (Runner not live yet).
HRD-092	task	middle	commons/cli/ shared CLI scaffold (Wave 2) — NewRootCmd + VersionCmd + Middleware hook) + StubCmd + healthz/readyz/metrics handlers. Vendored as §11.4.74 catalogue-check no-match. As-built evidence: §44.M of spec V3 r8 + c checks/HRD-092-commons-cli.md.
HRD-093	task	middle	sherald flavor scaffold — System Herald binary serving :24793 with cli.ServeC (HRD-033/034/040/046/056) + /v1/safety_state 501 stub → HRD-098.
HRD-094	task	middle	cherald flavor scaffold — Constitution Herald binary serving :24792 with cli.Se stubs + /v1/compliance 501 stub → HRD-028.
HRD-095	task	low	bherald flavor scaffold — Build Herald CLI-only binary with 3 §43 stubs (HRD
HRD-096	task	low	rherald flavor scaffold — Release Herald CLI-only binary with 3 §43 stubs (HR
HRD-097	task	low	iherald + scherald flavor scaffolds — Incident Herald serving :24794 with /v1/w HRD-024 + Scheduled-audit Herald CLI-only with 1 §43 stub (HRD-047). Pair has zero §43 stubs.
HRD-012	task	middle	Claude Code dispatcher live integration — claude --resume <UUID> - <envelope> exec + <<<HERALD-REPLY>>> JSON parse + claude_cod persistence per §33. §107 evidence: Plan 2 Task 6 commit 702b5a3 (live PASS claude CLI round-trip, structured reply parsed, Outcome+Summary non-empty) 4718c0e (live PASS 36.23s — session_state upsert under HeraldSystemTenant equality assertions on session_uuid + anchor_path + last_response JSONB round HRD-085..HRD-090 stay open for upstream-defined TaskRepository methods n Dispatch+session hot path.
HRD-010	task	middle	commons_storage live wiring — pgx pool + RLS-enforcing WithTenantContext fixed RLS-bypass bug via E14 falsifiability) + 9 migrations + background queue (digital.vasic.background bound via pgxTaskRepository) + Redis ACL (digital.v

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			pherald migrate up/status/down/validate subcommand + 3 new §107 e2e invariants + HRD-085..090 registered for queue-repository stubs
HRD-080	task	low	Refine I6 inheritance-gate invariant from blanket ".gitmodules-forbidden to constitution/entry in ".gitmodules" — paired meta-test test_i6_refinement_meta.sh with 3 anti-bluff subtests. Enables Foundation Helix-stack submodule installs.
HRD-017	task	middle	Propagate Universal §11.4.73 (main-spec versioning + revision discipline) and §11.4.74 (submodule-catalogue-first discovery) into parent constitution Constitution.md + AGENTS.md
HRD-014	task	middle	pherald CLI scaffold — Cobra root + version + 5 stubbed subcommands; pherald --json returns canonical build info
HRD-013	task	middle	commons_messaging + null:// adapter — full §11.0 Channel contract impl with fail_rate/latency/ceiling URL params + 8-case unit test suite
HRD-009b	task	low	commons_prefix module — §8.2 deterministic 3-letter prefix algorithm with Collision-resolution via fnv1a32 + table-driven tests
HRD-009	task	middle	commons module — full §11.0 Go type contract (Channel + Capabilities + Delivery + OutboundMessage + Body + Attachment + Recipient + Action + Priority + Recipient + InboundHandler + InboundEvent + Subscriber + SubscriberAlias + CloudEvent + TraceContext + Branding + ChannelID + PreferenceSet + ...) + Clock abstraction helper + DefaultBranding factory
HRD-007	task	middle	V3 r3 cross-doc sync + tracking-doc scaffold (Issues/Fixed/Status/Status_Summary/Issues_Summary/Fixed_Summary/CONTINUATION)
HRD-006	task	middle	V3 r2 flavor refinement — 9 flavors × per-channel interaction tables
HRD-005	task	middle	V3 r1 operator-product layer (§31..§36 + §18.2 expansion)
HRD-004	task	middle	V2 r3 — Go type contract closure + operational ops detail
HRD-003	task	middle	V2 r2 — close prose definition gaps + add operational guidance
HRD-002	task	middle	V2 r1 — architectural authoring (CloudEvents/Watermill/OTel/RLS/SLSA/9 flavors)
HRD-001	task	middle	V1 — initial MVP specification + Review section + Recommendations